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PROGRAMMING WITH C++ WK4

1.Scenario Analysis: Price Calculator

A price calculator is a simple Input/Output (I/O) system that helps users calculate the total cost of products they want to purchase. The user provides input such as the product price and the quantity they want to buy. The program receives these values and processes them by multiplying the price by the quantity. For example, if one product costs M50 and the customer wants to buy four items, the calculator multiplies M50 by 4 to produce a total of M200. The calculated amount is then displayed as output on the computer screen. The system can also include additional inputs such as discounts or tax rates. This makes the calculator useful in shops, businesses, and online stores. The main purpose of this I/O scenario is to demonstrate how information is entered into a computer, processed using a calculation, and returned to the user as meaningful output. It provides a simple example of how I/O works in programming.

2.Concept Research: Input Validation

Input validation is the process of checking information entered by a user before a computer program processes it. Its main role is to ensure that the input is correct, appropriate, and in the expected format. For example, a price calculator should make sure that the user enters a valid number instead of letters or negative values. Input validation helps prevent errors, incorrect calculations, and unexpected program behaviour. It also improves security by reducing the possibility of harmful or unexpected data being entered. In programming, validation can include checking data types, ranges, required fields, and acceptable formats. Therefore, input validation makes programs more reliable, accurate, secure, and user-friendly.

3.TOOL PRACTICE

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    string name;
```

```
    double price;
```

```
int quantity;
```

```
cout << "Enter your name: ";
```

```
cin >> name;
```

```
cout << "Enter the price of the product: ";
```

```
cin >> price;
```

```
cout << "Enter the quantity: ";
```

```
cin >> quantity;
```

```
double total = price * quantity;
```

```
cout << "\nCustomer: " << name << endl;
```

```
cout << "Total cost: M" << total << endl;
```

```
return 0;
```

```
}
```

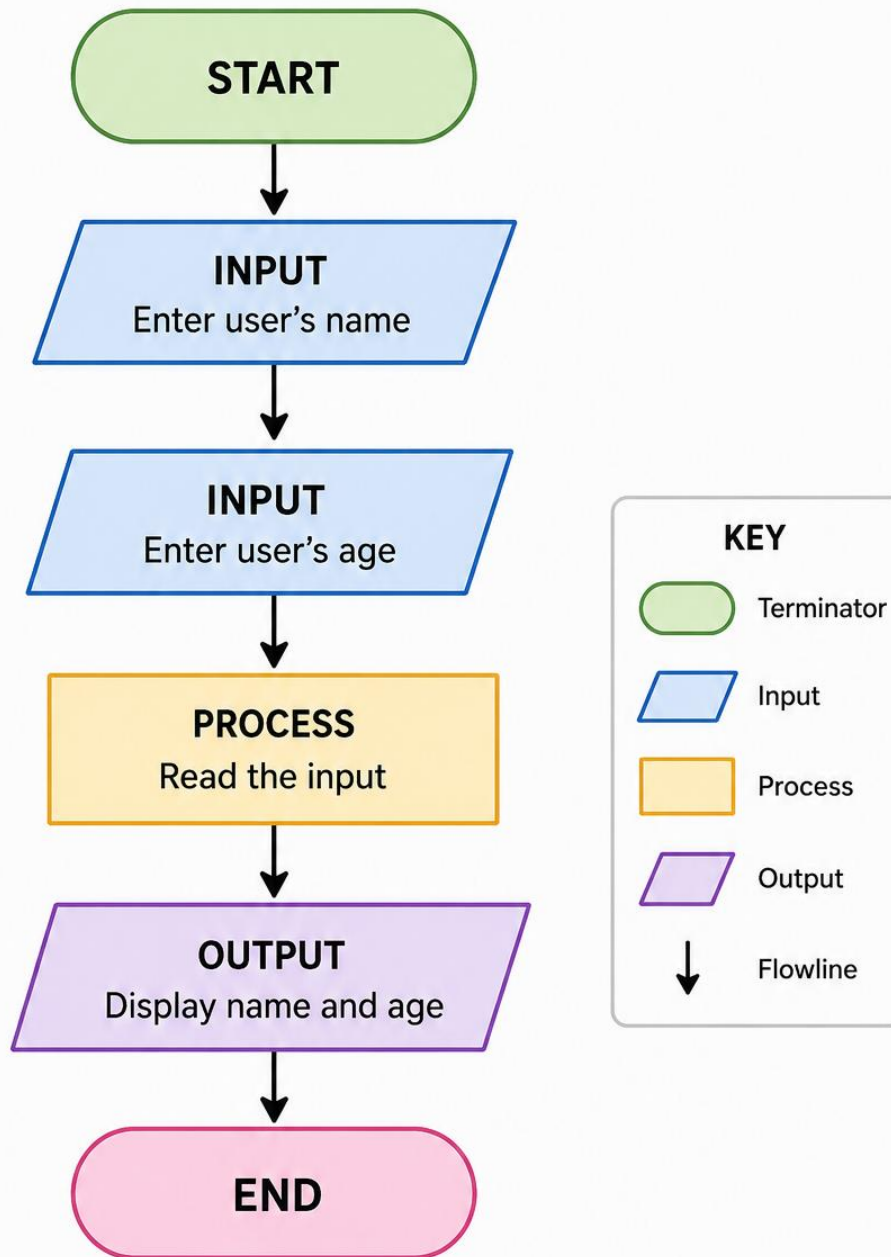
Reflection:

Writing the C++ I/O program in Visual Studio Code helped me understand how input and output work in programming. I used `cin` to allow the user to enter their name, product price, and quantity. I then used `cout` to display the information and the calculated total cost. The program also helped me understand how variables store different types of data, such as strings, integers, and decimal numbers. Using a simple multiplication calculation showed me how input can be processed to produce useful output. I learned that clear prompts are important because they guide users when

entering information. Overall, this practical activity improved my confidence in writing and running basic C++ programs.

4.FLOWCHAT DIAGRAM:

FLOWCHART FOR AN I/O PROGRAM



This flowchart shows the sequence of an I/O program where the user enters data (input), the program processes it, and then displays the result (output).

This flowchart represents a simple C++ input and output (I/O) program. The process begins with the Start symbol, which shows where the program begins execution. The user is then asked to enter their name and age. These values are received by the program as input using C++ input statements such as `cin`. After collecting the information, the program processes the input and prepares it for display. The name and age are then shown to the user using an output statement such as `cout`. Finally, the program reaches the End symbol, indicating that execution is complete. The flowchart makes the program's sequence easy to understand.